

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$(VII)' \quad \Psi_{\mathbf{D}} = \begin{pmatrix} a_1 + \mathbf{j}b_1 \\ a_2 + \mathbf{j}b_2 \\ a_3 + \mathbf{j}b_3 \\ a_4 + \mathbf{j}b_4 \end{pmatrix} \leftrightarrow \Psi_{\mathbf{C}}^{SR} = \begin{pmatrix} -b_1 + \mathbf{i}b_3 \\ -b_2 + \mathbf{i}b_4 \\ -a_1 + \mathbf{i}a_3 \\ -a_2 + \mathbf{i}a_4 \end{pmatrix}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ll} A) & J_1, J_2, J_3 \text{ unrestricted,} \quad l \geq 6 + \frac{1}{2}(J_1 + 2J_2 + 3J_3) + 2(a_1 + a_2) \\ B) & J_3 = 0, \quad l = 4 + \frac{1}{2}(J_1 + 2J_2) + 2(a_1 + a_2) \\ C) & J_2 = J_3 = 0, \quad l = 2 + \frac{1}{2}J_1 + 2(a_1 + a_2) \\ D) & J_1 = J_2 = J_3 = 0, \quad l = 2(a_1 + a_2) \end{array}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} W(\lambda_1, \lambda_2, \lambda_3) &= \Phi(\lambda_1, \lambda_2, \lambda_3, (\lambda_2\lambda_3), (\lambda_1\lambda_3), (\lambda_1\lambda_2), (\lambda_1\lambda_2\lambda_3)) \\ &= \Psi(\lambda_1 + \lambda_2 + \lambda_3, (\lambda_2\lambda_3) + (\lambda_1\lambda_3) + (\lambda_1\lambda_2), (\lambda_1\lambda_2\lambda_3)) \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$r(q_1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1+\beta t_j}{1+\beta t_{j'}} & 0 \\ 0 & 1 & q_1^{-1} \frac{t_j - t_{j'}}{1+\beta t_{j'}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \text{End}[\mathbb{Z}\llbracket q_1, q_1^{-1} \rrbracket \otimes V(t_j) \otimes V(t_{j'})]$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{1}{8\pi} \delta A &= \left\{ r_+ + M \left(1 + (r_+ - M)^{-1} \left(\frac{M^2 + a^2}{M} \right) \right) \right\} \delta M - \frac{a}{(r_+ - M)} \delta J \\ &= \frac{r_+^2 + a^2}{r_+ - M} \delta M - \frac{a}{r_+ - M} \delta J = \frac{1}{\kappa} \delta M - \frac{\Omega_H}{\kappa} \delta J, \end{aligned}$$